

Theme Park · Amenities — Functions Calling Functions — English Worksheet (Advanced)

Topic: Amenities — Functions Calling Functions · **Course:** Theme Park Creator · **Time:** about 50 minutes

Your park has a market row. Now it needs **amenities** — a **bathroom**, a **janitor room**.

Both are rooms, so **one** `build_room` helper builds both. Each amenity function calls the helper, then adds its own fitting. Then one more function calls **both amenities** — a chain three levels deep.

These are the blocks you use this lesson:

```
blocks.fill(QUARTZ_BLOCK, pos(x1, 0, z1), pos(x2, 3, z2), FillOperation.HOLLOW)
blocks.fill(AIR, pos(x1 + 1, 1, z1), pos(x1 + 1, 2, z1), FillOperation.REPLACE)
blocks.place(CAULDRON, pos(x1 + 1, 1, z1 + 1))
```

`blocks.fill(material, corner1, corner2, ...)` fills a box between two corners.

`FillOperation.HOLLOW` = outer **shell** only — the middle is **cleared to air**.

`FillOperation.REPLACE` = the **whole solid** box. `blocks.place(block, pos(x, y, z))` puts **one** block in **one** spot. `CAULDRON` = the toilet. `BARREL` = the janitor's supplies.

1 · Meet the Call Chain

A **helper function** does one job. Functions above it **call** it. Functions above *those* call **them**.

```
build_amenities(2)
├─▶ build_bathroom(2, 8, 0, 6)
│   ├──▶ build_room(QUARTZ_BLOCK, STONE, 2, 8, 0, 6)
│   └─▶ place CAULDRON
└─▶ build_janitor_room(10, 14, 0, 4)
    ├──▶ build_room(OAK_PLANKS, STONE, 10, 14, 0, 4)
    └─▶ place BARREL
```

One `build_room` runs twice — different materials, different corners. The helper takes **four corners**, so the caller decides where each room sits and how big it is.

```
def build_room(wall, floor, x1, x2, z1, z2):  
    blocks.fill(wall, pos(x1, 0, z1), pos(x2, 3, z2), FillOperation.HOLLOW)  
    blocks.fill(floor, pos(x1, 0, z1), pos(x2, 0, z2), FillOperation.REPLACE)  
    blocks.fill(AIR, pos(x1 + 1, 1, z1), pos(x1 + 1, 2, z1), FillOperation.REPLACE)
```

One helper is called by two functions, which are called by a third. What would you have to change if the doorway needed to be two blocks wide?

2 · Predict 🧠

Read each function. Write what you think happens.

```
def build_bathroom(x1, x2, z1, z2):
    build_room(QUARTZ_BLOCK, STONE, x1, x2, z1, z2)
    blocks.place(CAULDRON, pos(x1 + 1, 1, z1 + 1))

def build_janitor_room(x1, x2, z1, z2):
    build_room(OAK_PLANKS, STONE, x1, x2, z1, z2)
    blocks.place(BARREL, pos(x1 + 1, 1, z1 + 1))

build_bathroom(2, 8, 0, 6)
```

Which lines run, and in what order? Which materials reach `build_room`, and which never do?


```
def build_amenities(x):
    build_bathroom(x, x + 6, 0, 6)
    build_janitor_room(x + 8, x + 12, 0, 4)

build_amenities(2)
build_amenities(20)
```

`build_amenities` is called twice. How many rooms are built, and how many times does `build_room` run?

3 · Spot the Bug 🐛

Each block is broken. Read what it should do, rewrite it so it works, then explain why the original was wrong.

Bug A — This should build a bathroom **next to** a janitor room. Instead only **one** room is there — wooden, with a barrel. The cauldron is gone.

```
def build_amenities(x):  
    build_bathroom(x, x + 6, 0, 6)  
    build_janitor_room(x, x + 6, 0, 6)  
  
build_amenities(2)
```

Hint: look at the corners each amenity is given.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The bathroom should hold a cauldron. The room builds fine, but the **cauldron never survives**.

```
def build_bathroom(x1, x2, z1, z2):  
    blocks.place(CAULDRON, pos(x1 + 1, 1, z1 + 1))  
    build_room(QUARTZ_BLOCK, STONE, x1, x2, z1, z2)  
  
build_bathroom(2, 8, 0, 6)
```

Hint: `FillOperation.HOLLOW` clears the middle of the room to air. The cauldron is in the middle.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The janitor room should be a wide, shallow box. It comes out the **wrong shape**, and the barrel is not inside it.

```
def build_janitor_room(x1, x2, z1, z2):  
    build_room(OAK_PLANKS, STONE, x1, z1, x2, z2)  
    blocks.place(BARREL, pos(x1 + 1, 1, z1 + 1))  
  
build_janitor_room(10, 14, 0, 4)
```

Hint: the helper's parameters are `(wall, floor, x1, x2, z1, z2)`. Which value landed in `x2`?

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Build the Amenity Block 🚻

In your homework world, write **one** helper and **three** functions that call down into it:

```
def build_room(wall, floor, x1, x2, z1, z2):  
def build_bathroom(x1, x2, z1, z2):  
def build_janitor_room(x1, x2, z1, z2):  
def build_amenities(x):
```

`build_room` must build:

- **walls** from the `wall` material with `FillOperation.HOLLOW`
- a **floor** from the `floor` material
- a **doorway** carved with `AIR` so a visitor can walk in

`build_bathroom` calls `build_room`, then places a `CAULDRON` inside. `build_janitor_room` calls `build_room` with different materials, then places a `BARREL` inside. `build_amenities(x)` calls both, giving each its own corners so they stand side by side.

Only `build_room` may contain `blocks.fill`. Where the rooms sit must come from the corner parameters — no fixed corner numbers inside the helper.

Then build two amenity blocks in different corners of your park:

```
build_amenities(2)  
build_amenities(20)
```

Write your helper, your amenity functions, and the lines that call them:

5 · Show Your Work 📷 🎥

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.